Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a continuous source X, whose p.d.f. is given by

$$p_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0 \tag{1}$$

   Set $\lambda = 1$ and consider the following sequence of symbols

$$\{0.1, 0.0, 0.5, 2, 0.2\} \tag{2}$$

   (a) Design a quantizer based on compander-expander. Use 2 bits to represent each symbol. Show the equivalent non-uniform quantizer in the original signal domain.

   (b) Quantize the sequence of symbols using the quantizer above. Compute the reconstructed sequence at the decoder and the MSE distortion.

   (c) Quantize the sequence of symbols using a uniform scalar quantizer in the range $[0, 2]$. Compute the reconstructed sequence at the decoder and the MSE distortion.

   Solution

   (a) In order to design the nonlinear transformation used in the compander-expander quantizer, we need to write the following expression, which denotes the c.d.f. of the source X

$$y = T(x) = \int_{-\infty}^{x} p_X(a)da = \int_{0}^{x} \lambda e^{-\lambda a} da = \left[-e^{-\lambda a}\right]_0^x = 1 - e^{-\lambda x} \tag{3}$$

   The output y is quantized with a uniform quantizer whose intervals and reconstruction levels are

$$[0, 1/4) \rightarrow y_1 = 1/8 \tag{4}$$
$$[1/4, 1/2) \rightarrow y_2 = 3/8 \tag{5}$$
$$[1/2, 3/4) \rightarrow y_3 = 5/8 \tag{6}$$
$$[3/4, 1) \rightarrow y_4 = 7/8 \tag{7}$$

   The equivalent non-uniform quantizer in the original signal domain is obtained by inverting $T(x)$. That is,

$$x = T^{-1}(y) = -\frac{\log(1-y)}{\lambda} = \log \frac{1}{1-y} \tag{8}$$

   The output y is quantized with a uniform quantizer whose intervals and reconstruction levels are

$$[0, \log(4/3)) \rightarrow y_1 = \log(8/7) \tag{9}$$
$$[\log(4/3), \log(2)) \rightarrow y_2 = \log(8/5) \tag{10}$$
$$[\log(2), \log(4)) \rightarrow y_3 = \log(8/3) \tag{11}$$
$$[\log(4), +\infty) \rightarrow y_4 = \log(8) \tag{12}$$

(b) Quantization proceeds as follows

x	0.1	0.0	0.5	2	0.2
y	0.09	0.0	0.39	0.86	0.18
I_k	1	1	2	4	1
y_k	1/8	1/8	3/8	7/8	1/8
x_k	log(8/7)	log(8/7)	log(8/5)	log(8)	log(8/7)
e	-0.03	0.13	0.03	-0.08	0.07

The MSE is

$$MSE = \frac{1}{5}\sum_{j=1}^{5} e_j^2 = 0.006 \tag{13}$$

(c) A uniform quantizer in $[0, 2]$ is defined as follows

$$[0, 1/2) \rightarrow y_1 = 1/4 \tag{14}$$
$$[1/2, 1) \rightarrow y_2 = 3/4 \tag{15}$$
$$[1, 3/2) \rightarrow y_3 = 5/4 \tag{16}$$
$$[3/2, 2) \rightarrow y_4 = 7/4 \tag{17}$$

Quantization proceeds as follows

x	0.1	0.0	0.5	2	0.2
I_k	1	1	2	4	1
x_k	1/4	1/4	3/4	7/4	1/4
e	-0.15	-0.25	-0.25	0.25	-0.05

The MSE is

$$MSE = \frac{1}{5}\sum_{j=1}^{5} e_j^2 = 0.0425 \tag{18}$$

2. **Question (5 pts)**

Discuss the properties of speech signals, both in the time and in the frequency domain.

Solution

See slide set `MMSP_2.1_Speech coding`.

3. **Question (5 pts)**

(a) Illustrate the block diagram of a transform coder.

(b) Discuss the design of the optimal transform and quantizer based on the statistics of the input source.

Solution

See slide set `MMSP_1.2_Lossy_Coding`.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Let $R(D)$ denote the rate-distortion function of a discrete signal without memory.

   - [T] [F] $R(0) = \sigma_x^2$
     FALSE. The rate spent in case of lossless coding ($D = 0$) is lower bounded by the entropy of the source $H(X)$
   - [T] [F] $R(D_1) > R(D_2)$, if $D_1 > D_2$
     FALSE. The rate-distortion function is monotone decreasing.
   - [T] [F] $R_{TC}(D) < R_{PCM}(D)$, where $R_{TC}(D)$ and $R_{PCM}(D)$ are, respectively, the operational rate-distortion curves of transform coding and scalar quantization.
     FALSE. In general, $R_{TC}(D) \leq R_{PCM}(D)$. However, since the source is memoryless, transform coding does not lead to any coding gain, hence $R_{TC}(D) = R_{PCM}(D)$ (Note that in this case the KLT transform is the identity matrix).
   - [T] [F] If $x \in \mathcal{U}(0,1)$, $D(R) = \frac{1}{12}2^{-2R}$
     TRUE. The variance of a uniformly distributed random variable in $[0,1]$ is equal to $1/12$.

   (b) Consider an image coding system

   - [T] [F] Entropy coding of transform coefficients is invertible.
     TRUE. Entropy coding is lossless, thus invertible.
   - [T] [F] The coding efficiency decreases when the block size is increased.
     FALSE. Increasing the block size it is possible to capture inter-pixel correlations at longer distance (at the cost of higher computational complexity)
   - [T] [F] The coding efficiency of zonal quantization is higher than that of threshold quantization
     FALSE. Zonal quantization in unable to address the non-stationary behaviour of natural images.

   (c) Consider a video coding system.

   - [T] [F] The encoder computes a copy of the video frames identical to that available at the decoder.
     TRUE. Video coding adopts a DPCM closed-loop scheme. To avoid drift, the encoder and the decoder need to work on exactly the same signals.
   - [T] [F] The average PSNR of prediction residuals of full search motion estimation is smaller than (or equal to) that of fast search.
     FALSE. Fast search avoids to visit all motion vector candidates. Hence, MSE is smaller and PSNR higher.
   - [T] [F] Motion vectors are lossy coded.
     FALSE. Motion vector are lossless coded to ensure that both the encoder and the decoder compute exactly the same motion-compensated frame.

5. **Question (Matlab - 10pts)**

   A wireless sensor network is designed to perform speech acquisition. Each sensor records speech samples from the environment, encodes and transmits them to a central controller, where speech is analyzed. Assuming that:

   - To maximize recognition accuracy, the distortion introduced by lossy coding (SNR) should be greater than 15 dB.
   - Each sensor consumes 20 nJ to transmit 1 bit of data.

Compute the energy dissipated to transmit the first 2 seconds of the file *speech.wav* under the accuracy constraint, when a sensor uses:

(a) PCM coding.

(b) DPCM coding.

(c) PCM coding + Entropy coding (in this case assume to use exactly $H(X)$ bits per sample, where X denotes the source of quantized speech samples (assume a discrete memoryless source).

Solution

```
clear all;
close all;
clc;

%load speech
[x fs Nbits] = wavread('speech.wav');

%select first two seconds
x = x(1:2*fs);

R = 1:8;

Ebit = 20*10^-9;

SNR = NaN*ones(size(R));
SNR_dpcm = NaN*ones(size(R));
H = NaN*ones(size(R));
x_q = NaN*ones(size(x));
x_q_dpcm = NaN*ones(size(x));

%PCM quantization + Entropy coding
lb = -1;
ub = 1;
for r = R
    step = (ub-lb)/2^r;
    x_q = step.*round(x/step);
    h = hist(x_q,2^r);
    p = h./sum(h);
    p = p(p~=0);
    H(r) = -sum(p.*log2(p));
    SNR(r) = 10*log10(var(x)/(var(x-x_q)));

end
idx = find(SNR>15);
display(['Rate required with PCM: ' num2str(R(idx(1))) ' bits per sample']);
display(['Energy dissipated with PCM: ' num2str(R(idx(1))*2*fs*Ebit) ' Joules']);
display(['Rate required with PCM + EC: ' num2str(H(idx(1))) ' bits per sample']);
display(['Energy dissipated with PCM + EC: ' num2str(H(idx(1))*2*fs*Ebit) ' Joules']);
plot(R,SNR);
hold on
plot(H,SNR,'--b');

%DPCM quantization
error = x(1:end-1) - x(2:end);
ub = max(error);
lb = min(error);
for r = R
    step = (ub-lb)/2^r;
    x_q_dpcm(1) = step.*round(x(1)/step);
    for i=2:length(x)
        e = x(i)-x_q_dpcm(i-1);
        eq = step*round(e/step);
        x_q_dpcm(i) = x_q_dpcm(i-1) + eq;
    end
    SNR_dpcm(r) = 10*log10(var(x)/(var(x-x_q_dpcm)));
end
idx = find(SNR_dpcm>15);
display(['Rate required with DPCM: ' num2str(R(idx(1))) ' bits per sample']);
display(['Energy dissipated with DPCM: ' num2str(H(idx(1))*2*fs*Ebit) ' Joules']);
plot(R,SNR_dpcm,'r');
legend('PCM','PCM + EC','DPCM','location','northwest');
xlabel('Rate [bit/sample]');
ylabel('SNR [db]');
```